Problem I. All the Way Left

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2.5 seconds
Memory limit:	1024 megabytes

Little Drink Congee is a good friend of Little Cyan Fish and a member of the famous acting group *All the Way Left*. He has recently been practicing his ability to recognize directions on a stage. To practice, he has selected n **distinct** points $A_1, A_2, \cdots, A_n$ on the stage. The stage is represented as a two-dimensional Cartesian plane, where the i-th point is located at the coordinates (x_i, y_i). Little Drink Congee aims to travel through all these points in the order of $p_1, p_2, \cdots, p_n$. A traversal is a permutation p of length n, where each point A_{p_i} is connected to $A_{p_{i+1}}$ with an oriented line segment.

Little Drink Congee thinks a traversal is considered good if and only if the following condition holds:

- It is non-self-intersecting, i.e. no two segments intersect except for two adjacent segments intersecting at one common endpoint.
- For each $1 \le i \le n-2$, the i-th turn is left (or going straight). Formally, the cross product of $\overrightarrow{A_{p_i}A_{p_{i+1}}}$ and $\overrightarrow{A_{p_{i+1}}A_{p_{i+2}}}$ is non-negative.

Little Drink Congee wants to know the number of good traversals, modulo $(10^9 + 7)$. However, he needs to spend time with Little Cyan Fish and cannot solve this challenge himself. Please help him calculate it!

Input

The first line contains a single integer T $(1 \le T \le 10^4)$, denoting the number of test cases.

For each test case, the first line contains a single integer n $(1 \le n \le 2 \times 10^3)$.

In the next n lines, the i-th line contains two integers x_i and y_i $(1 \le x_i, y_i \le 10^9)$, denoting the coordinates of A_i. The coordinates of all the points are distinct.

It is guaranteed that the sum of n^2 over all test cases does not exceed 4×10^6.

Output

For each test case, output one line, containing the number of good traversals modulo $(10^9 + 7)$.

Example

standard input	standard output
3 4 1 1 3 1 2 2 2 3 3 1 1 1 2 1 3 6 1 1 2 1 2 2 2 3 3 2 4 2	6 2 13